Form-2

FINAL YEAR PROJECT

[PROPOSAL]

Project Title	EARLY DETECTION OF MALICIOUS OWNERSHIP TRANSACTIONS IN BROWSER EXTENSIONS VIA TEMPORAL BEHAVIOR DRIFT MODELING

DESCRIPTION

Browser extensions provide valuable functionality to enhance user experience, but they also pose significant security risks. Malicious actors can exploit extensions by performing unauthorized ownership transactions, injecting malware, or escalating privileges without users' knowledge. Traditional security mechanisms, such as signature-based detection or manual reviews, are limited in identifying these sophisticated, evolving threats. This project aims to develop a system that detects malicious ownership transactions early by modeling the temporal behavioral drift of browser extensions.

The proposed approach involves monitoring the behavioral patterns of browser extensions over time, including permission changes, network requests, file operations, and API interactions. Temporal modeling techniques, such as sequence analysis and time-series machine learning models, will be used to learn normal behavior for each extension. Deviations from these patterns—behavioral drift—can indicate potential malicious activity. By detecting such anomalies early, the system can prevent unauthorized changes and alert users or administrators before any significant damage occurs.

The framework will be implemented in a modular fashion: data collection, behavior modeling, drift detection, and alert generation. Real-world extension data will be used for training and testing, ensuring the solution is practical and effective. Evaluation metrics will include detection accuracy, precision, recall, and false positive rate, benchmarked against existing approaches.

This project provides not only a detection mechanism but also a comprehensive understanding of how browser extensions' behaviors evolve over time, offering insights for developers and security analysts. The system will be lightweight, automated, and adaptable to new extensions and threat patterns, making it a proactive cybersecurity solution.

PRIMARY GOALS

- *Collect and monitor temporal behavioral data of browser extensions.*
- *Model normal extension behavior and detect significant behavioral drift.*
- *Implement an early alert system to flag suspicious ownership transactions.*
- *Evaluate system performance on real-world extensions with measurable metrics.*

- *Semester 7 (4 months): Literature review, data collection, and behavior modeling*
- *Semester 8 (4 months): Drift detection implementation, system integration, testing, and evaluation*

RESOURCES REQUIRED

Software: Python, TensorFlow/PyTorch, Selenium/WebDriver, Browser APIs
Hardware: Mid-range computing system or cloud GPU instance
Datasets: Public browser extension repositories, malware and threat databases

Students Name	Signature	Supervisor Signature
M.Umer Khan		
Saifullah Imran		